

Package ‘soundmeteR’

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Type Package

Title R package for calculating sound pressure and intensity levels

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Description

An R package for sound analysis using octave and one-third octave spectral analysis, with calibration support for obtaining Sound Pressure Level (SPL) and Sound Intensity Level (SIL).

URL <https://github.com/ConservaSom/soundmeteR>

BugReports <https://github.com/ConservaSom/soundmeteR/issues>

License GPL (>=3)

Depends R (>= 3.5.0), tuneR, seewave, dplyr, progress

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calibration	<i>Extract calibration factor from reference signal</i>
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Description

Function to compute the calibration factor from a reference signal, converting measured levels from decibels at full scale (dBFS) to absolute values (dB) for subsequent analyses.

Usage

```
calibration(  
  file = NULL,  
  channel = "left",  
  from = 0,  
  to = Inf,  
  leqsignal = NA,  
  weighting = "none",  
  ref = 20  
)
```

Arguments

file	Specifies audio file to be analyzed. It can be a single file name on the computer or a Wave object on the environment (by default NULL). Only ".wav" files are accepted.
channel	Character. Choose “left” or “right” channel (by default "left"). Argument passed to mono function from tuneR to extract the desired channel.
from	Numeric. The start time (in seconds) of the segment to analyze. It can also be relative to the end of the file (in negative values). See examples for details.
to	Numeric. The end time (in seconds) of the segment to analyze. Could also be relative to the beginning of the file (in negative values). See examples for details.
weighting	Character. Defines the weighting curve for the analysis, passed to the dBweight function. Accepted values are "A", "B", "C", "D", "ITU", and "none" (by default "none"). See dBweight for details. # @param leqsignal Numeric. Specifies the sound pressure level (in dB SPL) for the signal in the audio (by default NA).
ref	Numeric. The reference value for dB conversion. For sound in water, the common reference is 1 μPa, and for sound in air, it is 20 μPa (by default 20).

dBtoLinear	<i>Convert deciBels scales to linear</i>
------------	--

Description

Function to convert decibels (dB) to linear values (μPa).

Usage

```
dBtoLinear(x, factor="IL", ref=1)
```

Arguments

x	Numeric. A numeric vector or a numeric matrix with dB values.
factor	Character. Specify the factor to use for conversion. Use SPL (Sound Pressure Level) for amplitude-base data, with factor 20, or IL (Intensity Level) for power-base data, with factor 10 (by default "IL").
ref	Numeric. The reference value for dB conversion. For sound in water, the common reference is 1 μPa , and for sound in air, it is 20 μPa (by default 1).

Details

For further details on selecting the appropriate factor, we recommend consulting [this](#) web page.

Value

The same input object with the values converted.

See Also

[rms.dB](#), [LineartodB](#), [convSPL](#)

Examples

```
dBtoLinear(c(80,60,65,62))  
LineartodB(dBtoLinear(c(80,60,65,62)))
```

freq.bands

*Interval Limits***Description**

Function to compute frequency intervals based on patterns defined by the user. This function is adapted from the [octaves](#) functions from [seewave](#) package.

Usage

```
freq.bands(x, interval = 2, below = 3, above = 3)
```

Arguments

x	Numerical. Frequency used as the central point from which bands are calculated in kHz (by default 2).
interval	Numeric. Specifies the interval pattern to be applied. See details for further information.
below	Numerical. Number of intervals below x (by default 3).
above	Numerical. Number of intervals above x (by default 3).

Details

The specified interval is applied as $x/\text{interval}$ (for the lower bound) and $x \times \text{interval}$ (for the upper bound). Some examples of intervals that can be applied are:

- Octaves = 2
- Third of an octave = $2^{(1/3)}$
- Perfect fifth (music theory) = $3/2$
- Major third (music theory) = $5/4$
- Minor third (music theory) = $6/5$
- For additional values that can be used, please refer to this [link](#).

Value

A numeric vector containing the frequency limits of each interval.

See Also

[octaves](#)

Examples

```
freq.bands(1000, interval=2, below = 1, above = 1) #octaves
freq.bands(1000, interval=2^(1/3), below = 3, above = 3) #Third of octaves

#https://academics.hamilton.edu/music/spellman/class_notes/music_theory.htm
freq.bands(440, interval=3/2, below = 0, above = 1) #Perfect fifth (music theory) of A 440
freq.bands(440, interval=5/4, below = 0, above = 1) #Major third (music theory) of A 440
freq.bands(440, interval=6/5, below = 0, above = 1) #Minor third (music theory) of A 440
```

leqbands

Equivalent Level per octaves or one-third octaves

Description

Function to compute the Equivalent Level (Leq) across octaves or one-third octaves.

Usage

```
leqbands(
  files = "wd",
  channel = "left",
  from = 0,
  to = Inf,
  weighting = "none",
  bands = "thirds",
  ref = 20,
  Leq.calib = NULL,
  Calib.value = NULL,
  saveresults = F,
  outname = NULL,
  progressbar = T
)
```

Arguments

files	Specifies audio file(s) to be analyzed. It can be set to "wd" to select all ".wav" files in the work directory, a single file name, a character vector with multiple file names, a Wave object, or a list of Wave objects (by default "wd"). Only ".wav" files are accepted.
channel	Character. Choose "left" or "right" channel (by default "left"). Argument passed to mono function from tuneR to extract the desired channel.
from	Numeric. The start time (in seconds) of the segment to analyze. It can also be relative to the end of the file (in negative values). See examples for details.
to	Numeric. The end time (in seconds) of the segment to analyze. Could also be relative to the beginning of the file (in negative values). See examples for details.

weighting	Character. Defines the weighting curve for the analysis, passed to the <code>dBweight</code> function. Accepted values are "A", "B", "C", "D", "ITU", and "none" (by default "none"). See <code>dBweight</code> for details.
bands	Character. Use "octaves" for octave bands or "thirds" for one-third octave bands intervals (by default "thirds").
ref	Numeric. The reference value for dB conversion. For sound in water, the common reference is 1 μ Pa, and for sound in air, it is 20 μ Pa (by default 20).
Leq.calib	Numeric. Specifies the sound pressure level (in dB SPL) for the signal in the audio (by default NULL). Cannot be set if <code>Calib.value</code> is specified.
Calib.value	Numeric. Specifies the calibration value (by default NULL). Can not be set if <code>Leq.calib</code> is specified.
saveresults	Logical. Set to TRUE to save the results to a .txt file (by default FALSE).
outname	Character. If <code>saveresults</code> is TRUE, specifies a name to append to the file name in the txt file (by default NULL).
progressbar	Logical. Set TRUE to display a progress bar showing elapsed time and the last completed file number, or FALSE to hide it (default TRUE).

Details

Caution: Ensure that the audio file has a duration with whole second values to avoid bugs. For example: 35s, 60s, 19s. By default, the function will truncate your audio file to the next whole second.

This function only works with mono audio files, and the audio must have a sampling rate of at least 44,100 Hz.

If you are working with decibels at full scale (dBFS), we recommend setting `ref=1` so the results will be relative to 0 dBFS.

References

Power spectrum adapted from: Carcagno, S. 2013. Basic Sound Processing with R [Blog post]. Retrieved from <http://samcarcagno.altervista.org/blog/basic-sound-processing-r/>

Miyara, F. 2017. Software-Based Acoustical Measurements. Springer. 429 pp. DOI: 10.1007/978-3-319-55871-4

Examples

```
data(tham)
leqbands(tham)
leqbands(tham, Calib.value=130.24)
leqbands(tham, Calib.value=130.24, weighting="A")
leqbands(tham, Calib.value=130.24, weighting="A", bands="octaves")

leqbands(tham, Leq.calib=48.17, weighting="A")

leqbands(tham, from=-3.49, to=-0.9, ref=1) #dB at Full Scale
leqbands(tham, from=-3.49, to=-0.9, Calib.value=130.24) #song Sound Pressure Level
leqbands(tham, from=0, to=3.8, Calib.value=130.24) #background Sound Pressure Level
```

leqbands_cal

*leqbands analysis for audiofiles with reference signal***Description**

This function passes the parameters to [leqbands](#) to automate the calibration process and return spectral analysis results in dB SPL.

Usage

```
leqbands_cal(
  files = "wd",
  channel = "left",
  from = 0,
  to = Inf,
  CalibPosition = NULL,
  CalibValue = NULL,
  ref = 20,
  weighting = "none",
  bands = "thirds",
  saveresults = F,
  outname = NULL,
  progressbar = T
)
```

Arguments

files	Specifies audio file(s) to be analyzed. It can be set to "wd" to select all ".wav" files in the work directory, a single file name, a character vector with multiple file names, a Wave object, or a list of Wave objects (by default "wd"). Only ".wav" files are accepted.
channel	Character. Choose "left" or "right" channel. Argument passed to mono function from tuneR (by default "left").
from	Numeric. The start time (in seconds) of the segment to analyze. It can also be relative to the end of the file (in negative values). See examples for details.
to	Numeric. The end time (in seconds) of the segment to analyze. It can also be relative to the beginning of the file (in negative values). See examples for details.
CalibPosition	Numeric. Specifies the calibration position. It can be a negative (relative to the sound file duration) or a positive value, or a data.frame containing these combinations. This parameter is used in conjunction with CalibValue.
CalibValue	Numeric. Specifies the calibration value. If CalibPosition is provided, it serves as the reference value. If CalibPosition is absent, CalibValue is used as the calibration reference.

ref	Numeric. The reference value for dB conversion. For sound in water, the common reference is 1 μ Pa, and for sound in air,it is 20 μ Pa (by default 20).
weighting	Character. Defines the weighting curve for the analysis, passed to the dBweight function. Accepted values are "A", "B", "C", "D", "ITU", and "none" (by default "none"). See dBweight for details.
bands	Character. Use "octaves" for octave bands or "thirds" for one-third octave bands intervals (by default "thirds").
saveresults	Logical. Set to TRUE to save the results to a .txt file (by default FALSE).
outname	Character. If saveresults is TRUE, specifies a name to append to the file name in the txt file (by default NULL).
progressbar	Logical. Set TRUE to display a progress bar showing elapsed time and the last completed file number, or FALSE to hide it (default TRUE).

See Also

[leqbands](#)

LineartodB	<i>Convert linear scales to deciBels</i>
------------	--

Description

Function to convert linear values (μ Pa) to decibels (dB).

Usage

LineartodB(x, factor="IL", ref=1)

Arguments

x	Numeric. A vector or matrix containing linear values (in μ Pa).
factor	Character. Specify the factor for conversion. Use SPL (Sound Pressure Level) for amplitude-base data, with factor 20, or IL (Intensity Level) for power-base data, with factor 10 (by default IL.)
ref	Numeric. The reference value for dB conversion. For sound in water, the common reference is 1 μ Pa, and for sound in air,it is 20 μ Pa (by default 1).

Details

For details about the factor choice, we recommend consulting [this](#) web page.

Value

The same input object with the values converted.

See Also

[rms.dB](#), [dBtoLinear](#), [convSPL](#)

Examples

```
dBtoLinear(c(80,60,65,62))
LineartodB(dBtoLinear(c(80,60,65,62)))
```

pwrspec

Power Spectrum of a sound file

Description

This functions computes a power spectrum from a sound file.

Usage

```
pwrspec(
  file,
  channel = "left",
  from = 0,
  to = Inf,
  bandpass = c(0, Inf),
  res.scale = "microPa",
  ref = 1
)
```

Arguments

file	A wave file path or a class Wave object already loaded into R environment.
channel	Character. Choose “left” or “right” channel. Argument passed to mono function from tuneR (by default "left").
from	Numeric. The start time (in seconds) of the segment to analyze. It can also be relative to the end of the file (in negative values). See examples for details.
to	Numeric. The end time (in seconds) of the segment to analyze. It can also be relative to the beginning of the file (in negative values). See examples for details.
bandpass	A vector of length two specifying the lower and upper limits of the bandpass interval in Hz.
res.scale	Character. Specifies the scale for adjusting the power spectrum amplitude. Use microPa for linear values in μPa , or dB for decibel values in dB SPL (by default microPa).
ref	Numeric. The reference value for dB conversion. For sound in water, the common reference is 1 μPa , and for sound in air, it is 20 μPa (by default 1).

Value

A data.frame with columns:

- Freq.Hz: numeric, frequency in Hertz
- Amp.microPa: numeric, amplitude in μPa (present when res.scale = "microPa")
- Amp.dB: numeric, amplitude in dB SPL (present when res.scale = "dB")

References

Power spectrum adapted from: Carcagno, S. 2013. Basic Sound Processing with R [Blog post]. Retrieved from <http://samcarcagno.altervista.org/blog/basic-sound-processing-r/>

Miyara, F. 2017. Software-Based Acoustical Measurements. Springer. 429 pp. DOI: 10.1007/978-3-319-55871-4

Examples

```
data(tham)
pwrspec(tham)

#with from, to and bandpass
pwrspec(tham, from=3.8, to=7.1, bandpass=c(900,2000))

#from and to relative to duration (negatives values from the end of the soundfiles)
pwrspec(tham, from=-3.49, to=-0.9, bandpass=c(900,2000))
```

rms.dB	<i>Root Mean Square with dB values</i>
--------	--

Description

Function to compute the root mean square (RMS) of values in decibels (dB).

Usage

```
rms.dB(x, level = "SPL", na.rm = FALSE)
```

Arguments

x	Numeric. A vector or a matrix with decibels values (dB).
level	Character. Specify the factor for conversion. Use SPL (Sound Pressure Level) for amplitude-base data, with factor 20, or IL (Intensity Level) for power-base data, with factor 10 (By default SPL).
na.rm	Logical. Argument passed to mean . If TRUE removes NA (By default FALSE).

Details

This function converts the decibels data to linear values using the [dBtoLinear](#) function, computes the Root Mean Square (rms), and then converts the result back to decibels using the [LineartodB](#) function.

This function was adapted from the [meandB](#) and [rms](#) functions from [seewave](#) package. See their documentation for more details.

Value

A numeric value that representing the root mean square of x.

See Also

[meandB](#), [rms](#)

Examples

```
rms.dB(c(80, 60, 65, 62))
```

song.level

RMS from a sample of a sound file

Description

RMS (Root Mean Square) from a sample of a sound file.

Usage

```
song.level(
  files = "wd",
  channel = "left",
  from = 0,
  to = Inf,
  freq.interval = c(0, Inf),
  fdom.int = c(0, Inf),
  wl = 512,
  ovlp = 50,
  CalibPosition = NULL,
  CalibValue = NULL,
  freq.weight = "none",
  ref = 20,
  progressbar = T
)
```

Arguments

files	Specifies audio file(s) to be analyzed. It can be set to "wd" to select all ".wav" files in the work directory, a single file name, a character vector with multiple file names, a Wave object, or a list of Wave objects (by default "wd"). Only ".wav" files are accepted.
channel	Character. Choose "left" or "right" channel. Argument passed to mono function from tuneR (by default "left").
from	Numeric. Specifies the start time (in seconds) of the segment to analyze. It can also be relative to the end of the file (in negative values). See examples for details.
to	Numeric. Specifies the end time (in seconds) of the segment to analyze. It can also be relative to the beginning of the file (in negative values). See examples for more details.
freq.interval	Specifies the frequency interval for computing the RMS. It can be a vector of length two, with the lower and upper frequency bounds (in Hz), or a pattern to calculate a interval (e.g., octaves). For more details, refer to freq.bands .
fdom.int	A vector of length two specifying the lower and upper frequency bounds (in Hz) to determine the dominant frequency. This frequency will be used as the center of the interval only if a pattern is specified in freq.interval.
wl	Numeric. Window length for the analysis. It must be a even number of points (by default 512).
ovlp	Numeric. Overlap percentage between two successive windows (by default 50). Argument passed to meanspec function from seewave .
CalibPosition	Numeric. Specifies the calibration position. It can be a negative (relative to the sound file duration) or a positive value, or a data.frame containing these combinations. This parameter is used in conjunction with CalibValue.
CalibValue	Numeric. Canbe a value to apply of the ref value from a calib signal (specified by CalibPosition).
freq.weight	Character. Defines the weighting curve for the analysis, passed to the dBweight function. Accepted values are "A", "B", "C", "D", "ITU", and "none" (by default "none"). See dBweight for details.
ref	Numeric. The reference value for dB conversion. For sound in water, the common reference is 1 μ Pa, and for sound in air, it is 20 μ Pa (by default 20).
progressbar	Logical. Set to TRUE to display a progress bar showing elapsed time and the last completed file number, or FALSE to hide it (default TRUE).

Examples

```

song.level(tham, freq.interval=c(22, 20000))

song.level(tham, freq.interval=c(22, 20000), CalibValue = 130.24)

song.level(tham, freq.interval=c(22, 20000), CalibValue = 130.24, freq.weight="A")

song.level(tham, freq.interval=c(22, 20000), CalibValue = 130.24, freq.weight="B")

```

```
song.level(tham, freq.interval=c(22, 20000), CalibValue = 130.24, freq.weight="C")

song.level(tham, from = 3.883035, to=7.044417, freq.interval=c(22.09429, 22627.38), CalibValue = 130.24, freq.weight="A")

#Perfect fifth (music theory)
song.level(tham, fdom.int = c(800, 2000), from = 3.8, to=7, freq.interval=3/2, CalibValue = 130.24, freq.weight="A")

#Perfect fifth (music theory)
song.level(tham, fdom.int = c(800, 2000), from = 3.8, to=7, freq.interval=3/2, CalibValue = 130.24, freq.weight="A")
```

soundmeter	<i>Sound meter measurements</i>
------------	---------------------------------

Description

Funtion that performs sound meter measurements.

Usage

```
soundmeter(
  files = "wd",
  channel = "left",
  from = 0,
  to = Inf,
  CalibPosition = NULL,
  CalibValue = NULL,
  ref = 20,
  fw = "none",
  bands = "octaves",
  tw = "fast",
  saveresults = F,
  outname = NULL,
  bandpass = c(0, Inf),
  progressbar = T
)
```

Arguments

files	Specifies audio file(s) to be analyzed. It can be set to "wd" to select all ".wav" files in the work directory, a single file name, a character vector with multiple file names, a Wave object, or a list of Wave objects (by default "wd"). Only ".wav" files are accepted.
channel	Character. Choose “left” or “right” channel. Argument passed to mono function from tuneR (by default "left").

from	Numeric. Specifies the start time (in seconds) of the segment to analyze. It can also be relative to the end of the file (in negative values). See examples for details.
to	Numeric. Specifies the end time (in seconds) of the segment to analyze. It can also be relative to the beginning of the file (in negative values). See examples for more details.
CalibPosition	Numeric. Specifies the calibration position. It can be a negative (relative to the sound file duration) or a positive value, or a data.frame containing these combinations. This parameter is used in conjunction with CalibValue.
CalibValue	Numeric. Specifies the calibration value (by default NULL). If "CalibPosition" is provided, it serves as the reference value. If "CalibPosition" is absent, "CalibValue" is used as the calibration reference.
ref	Numeric. The reference value for dB conversion. For sound in water, the common reference is 1 μ Pa, and for sound in air, it is 20 μ Pa (by default 20).
fw	Character. Defines the weighting curve for the analysis, passed to the dBweight function. Accepted values are "A", "B", "C", "D", "ITU", and "none" (by default "none"). See dBweight for details.
bands	Character. Use "octaves" for octave bands or "thirds" for one-third octave bands intervals (by default "thirds").
tw	Character. Specifies the time window to be used. Accepted values are "fast" and "slow" (default "fast").
saveresults	Logical. Set TRUE to save the results to a .txt file (by default FALSE).
outname	Character. If saveresults is TRUE, specifies a name to append to the file name in the txt file (by default NULL).
bandpass	Numeric. A vector of length two specifying the lower and upper limits of the bandpass interval in Hz.
progressbar	Logical. Set to TRUE to display a progress bar showing elapsed time and the last completed file number, or FALSE to hide it (default is TRUE).

Details

If your reference signal is in a separate file, we recommend obtaining the CalibValue using the [leqbands](#) function. Examples in the documentation provide further details.

Examples

```
data("tham")
soundmeter(tham, CalibValue = 130.24, tw = "slow") # slow time window with calib value
soundmeter(tham, CalibValue = 130.24, tw = "fast") # fast

soundmeter(tham, CalibValue = 130.24, tw = "fast", fw = "A") # fast with frequency weight

soundmeter(tham, CalibValue = NULL, tw = "fast", ref = 1) # fast time window in dBFS
```

sumdB	<i>Sum with dB values</i>
-------	---------------------------

Description

Sum with dB values

Usage

```
sumdB(x, level = "IL", na.rm = FALSE)
```

Arguments

x	Numerical. A vector or a matrix with decibels values (dB).
level	Character. Specify the factor for conversion. Use SPL (Sound Pressure Level) for amplitude-base data, with factor 20, or IL (Intensity Level) for power-base data, with factor 10 (By default IL).
na.rm	Logical. Argument passed to sum . If TRUE removes NA (by default FALSE).

Details

Function to compute the sum of decibels values (dB).

This function converts the decibels data to linear values using the [dBtoLinear](#) function, computes the sum, and then converts the result back to decibels using the [LineartodB](#) function.

Value

A numeric value representing the sum of x.

See Also

[moredB](#)

Examples

```
sumdB(c(80,60,65,62))
sumdB(c(30,30), level="IL")
sumdB(c(30,30), level="SPL")
```

tham

Thamnophilus stictocephalus song in a landscape

Description

Thamnophilus stictocephalus song in a landscape

Usage

```
data(tham)
```

Format

An object of class "Wave"; see [readWave](#).

Details

A landscape recording presenting *Thamnophilus stictocephalus* song between 3.883035 and 7.044417 seconds. The "calib.value" for this record is 130.24.

Source

Recording by Ingrid Maria Denóbile Torres

Tweighting

Time Weighting of a Audiofile

Description

This function split your audiofile into smaller segments, defined as fast (0.125s) and slow (1s), and analyze each one using the [leqbands](#) function.

Usage

```
Tweighting(
  file,
  window = "fast",
  channel = "left",
  weighting = "none",
  bands = "thirds",
  ref = 20,
  Calib.value = NULL,
  bandpass = c(0, Inf),
  from = 0,
  to = Inf
)
```


Arguments

file	Wave object
window	Character. Specifies the time window to be used. Accepted values are fast and slow (default fast).
channel	Character. Choose “left” or “right” channel. Argument passed to mono function from tuneR (by default "left").
weighting	Character. Specifies the end time (in seconds) of the segment to analyze. It can also be relative to the beginning of the file (in negative values). See examples for more details.
bands	Character. Use "octaves" for octave bands or "thirds" for one-third octave bands intervals (by default "thirds").
ref	Numeric. Defines the weighting curve for the analysis, passed to the dBweight function. Accepted values are "A", "B", "C", "D", "ITU", and "none" (by default "none"). See dBweight for details.
Calib.value	Numeric. Specifies the calibration value (by default NULL). Cannot be set if Leq.calib is specified.
bandpass	Numeric. A vector of length two specifying the lower and upper limits of the bandpass interval in Hz.
from	Numeric. Specifies the start time (in seconds) of the segment to analyze. It can also be relative to the end of the file (in negative values). See examples for details.
to	Numeric. Specifies the end time (in seconds) of the segment to analyze. It can also be relative to the beginning of the file (in negative values). See examples for more details.
Leq.calib	Numeric. Specifies the sound pressure level (in dB SPL) for the signal in the audio file (by default NULL). Cannot be set if Calib.value is specified.

See Also

[leqbands](#), [soundmeter](#)

Examples

```
# creating an example sound file
som <- sine(1000, duration = 44500)

# default options without calibration (results in dBFS)
Tweighting(som)

# Simulation of a calib signal with a Leq of 94dB in the field ####
# fast
Tweighting(som, window = "fast", bands = "octaves", Leq.calib = 94)
# slow
Tweighting(som, window = "slow", bands = "octaves", Leq.calib = 94)

# Using the result of the simulation above to calibrate the sound and output
```

```
# fast
Tweighting(som, window = "fast", bands = "octaves", Calib.value = 309.67)
# slow
Tweighting(som, window = "slow", bands = "octaves", Calib.value = 309.67)

# With tham data
data(tham)
Tweighting(tham, window = "fast", bands = "octaves", Calib.value = 130.24) # fast
Tweighting(tham, window = "slow", bands = "octaves", Calib.value = 130.24) # slow
```

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